

Design Calculation Sheet - VBF-T2R2-CALC-01

Case t2_r2 - Virtual Battery Factory - generated 2026-08-26

1. Inputs (final params)

Parameter	Value	Unit	Formula	Source
Positive electrode thickness	110.000 um	um	$t \times 1e6$	r3_vA_k2e15.json
Negative electrode thickness	150.000 um	um	$t \times 1e6$	r3_vA_k2e15.json
Separator thickness	10.000 um	um	$t \times 1e6$	r3_vA_k2e15.json
Positive / negative / separator porosity	0.30 / 0.42 / 0.55	-	-	r3_vA_k2e15.json
Positive / negative AM volume fraction	0.70 / 0.58	-	-	r3_vA_k2e15.json
Positive / negative particle radius	3.00 / 2.50 um	um	$r \times 1e6$	r3_vA_k2e15.json
Current collectors Al/Cu	8 / 6 um	um	$t \times 1e6$	r3_vA_k2e15.json
Electrolyte conductivity (constant override)	2.500	S/m	-	r3_vA_k2e15.json (estimate, annotation)
Electrolyte diffusivity	4.000e-10	m ² /s	-	r3_vA_k2e15.json
Cation transference number	0.350	-	-	r3_vA_k2e15.json
SEI kinetic rate constant (coating bridge)	2.00e-15	m/s	-	r3_vA_k2e15.json (estimate, annotation)
Positive / negative electrode density	3262 / 1657	kg/m ³	-	chen2020_dump.json
Electrode height / width	65 / 1580	mm	$x \times 1e3$	chen2020_dump.json
Voltage window	2.5 - 4.2	V	-	chen2020_dump.json
Nominal capacity	5.0	Ah	-	chen2020_dump.json

2. Capacity and energy

Parameter	Value	Unit	Formula	Source
1C discharge capacity (25 C, DFN)	6.908891	Ah	simulation integration	final_1c_dfn.json, capacity_ah
4C charge accepted before 4.2 V cutoff	0.958232	Ah	charge-segment time x 4C rate	final_4c_dfn.json, capacity_ah
1C discharge energy	25.351293	Wh	time integration of $V \times I$	final_calc.json, energy_wh
1C midpoint voltage	3.875127	V	energy/capacity midpoint	final_calc.json, midpoint_voltage_v
DC resistance	2.467664	mOhm	set value / pulse caliber	final_calc.json, dcr_ohm
Power density	35233.5724	W/kg	energy/mass caliber	final_calc.json, power_density_w_kg

3. Energy density

